

7. For the transfer function $H(z) = \frac{1}{z-0.5}$, determine whether the system is stable and find the corresponding impulse response sequence.

$$z - 0.5 = 0 \Rightarrow z = 0.5 \quad \text{pole} \quad |0.5| < 1 \Rightarrow \text{the system is stable}$$

$$H(z) = \frac{1}{z-0.5} = \frac{z^{-1}}{1-0.5z^{-1}} = z^{-1} \cdot \left(\frac{1}{1-0.5z^{-1}} \right) \Rightarrow h[n] = 0.5^{n-1} u[n-1]$$

$$z^{-1} \left\{ \frac{1}{1-0.5z^{-1}} \right\} = a^n u[n]$$

8. Study the stability of the system with transfer function

$$H(z) = \frac{1}{z^2 - z + 0.5}$$

and determine its impulse response.

$$z^2 - z + 0.5 = 0 \Rightarrow 2z^2 - 2z + 1 = 0 \quad \Delta = 4 - 4 \cdot 2 \cdot 1 = -4$$

$$z_{1,2} = \frac{2 \pm \sqrt{-4}}{4} = \frac{1 \pm \sqrt{-1}}{2} = 0.5 \pm 0.5j \quad \text{poles}$$

$$|z_{1,2}| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{1}{2}} < 1 \Rightarrow \text{the system is stable}$$

$$z^2 - 2r \cos(\omega_0) z + r^2 = z^2 - z + 0.5$$

$$r^2 = 0.5 \Rightarrow r = \sqrt{\frac{1}{2}}$$

$$-2r \cos(\omega_0) = 1 \Rightarrow -2 \frac{1}{\sqrt{2}} \cos(\omega_0) = -1 \Rightarrow \cos(\omega_0) = \frac{\sqrt{2}}{2} \Rightarrow \omega_0 = \frac{\pi}{4}$$

$$\mathcal{Z} \{ r^n \cos(\omega_0 n) u[n] \} = \frac{r \cos(\omega_0) z}{z^2 - 2r \cos(\omega_0) z + r^2} = \frac{\frac{1}{\sqrt{2}} \cdot \cos(\omega_0) \cdot z}{z^2 - z + 0.5} = \frac{\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \cdot z}{z^2 - z + 0.5}$$

$$H(z) = z^{-1} \cdot 2 \cdot \frac{0.5 z}{z^2 - z + 0.5} \quad n-1$$

$$\mathcal{Z}^{-1} \left\{ z^{-1} \cdot 2 \cdot \frac{0.5 z}{z^2 - z + 0.5} \right\} = 2 \cdot \left(\frac{1}{\sqrt{2}} \right) \cdot \cos\left(\frac{\pi}{4}(n-1)\right) u[n-1]$$

9. A processor has a transfer function $\frac{0.4}{z-0.5}$. Find the response to a discrete unit step (first four terms only).

$$H(z) = \frac{0.4}{z-0.5} = \frac{0.4 z^{-1}}{1-0.5 z^{-1}}$$

$$\frac{Y(z)}{X(z)} = \frac{0.4 z^{-1}}{1-0.5 z^{-1}} \Rightarrow Y(z) - 0.5 z^{-1} Y(z) = 0.4 z^{-1} X(z) \Rightarrow$$

$$\Rightarrow Y(z) - 0.5 Y(z) z^{-1} = 0.4 X(z) z^{-1} \Rightarrow Y[n] - 0.5 Y[n-1] = 0.4 X[n-1] \Rightarrow Y[n] = 0.4 X[n-1] + 0.5 Y[n-1] \quad \Bigg\} \Rightarrow$$

$$X[n] = u[n]$$

$$n=0 \quad Y[0] = 0.4 \cdot 0 + 0.5 \cdot 0 = 0$$

$$n=1 \quad Y[1] = 0.4 \cdot 1 + 0.5 \cdot 0 = 0.4$$

$$n=2 \quad Y[2] = 0.4 \cdot 1 + 0.5 \cdot 0.4 = 0.4 + 0.2 = 0.6$$

$$n=3 \quad Y[3] = 0.4 \cdot 1 + 0.5 \cdot 0.6 = 0.4 + 0.3 = 0.7$$

14. Given the transfer function $H(z) = \frac{z+1}{z-0.5}$, calculate the frequency response at 0 Hz, 1 Hz, and 2 Hz if the sampling frequency $f_s = 8$ Hz. Provide both gain and phase angle for each frequency.

$$z = e^{j\Omega} = e^{j 2\pi \frac{f}{f_s}}$$

$$f = 0 \text{ Hz} \Rightarrow \Omega = 2\pi \frac{0}{8} = 0 \Rightarrow z = e^{j \cdot 0} = 1$$

$$H(1) = \frac{1+1}{1-0.5} = \frac{2}{0.5} = 4$$

$$gain: |4| = 4$$

$$phase: 0^\circ$$

$$f = 1 \text{ Hz} \Rightarrow \Omega = 2\pi \frac{1}{8} = \frac{\pi}{4} \Rightarrow z = e^{j \frac{\pi}{4}} = \cos \frac{\pi}{4} + j \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2}$$

$$\text{numerator: } N = z + 1 = \frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2} + 1 = \frac{2 + \sqrt{2}}{2} + j \frac{\sqrt{2}}{2}$$

$$\text{magnitude: } \sqrt{\left(\frac{2 + \sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} = \sqrt{2 + \sqrt{2}}$$

$$\text{phase: } \arctan\left(\frac{\frac{\sqrt{2}}{2}}{\frac{2 + \sqrt{2}}{2}}\right) = \arctan\left(\frac{\sqrt{2}}{\sqrt{2} + 2}\right) = \frac{\pi}{8}$$

$$\text{denominator: } D = z - 0.5 = \frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2} - 0.5 = \frac{\sqrt{2} - 1}{2} + j \frac{\sqrt{2}}{2}$$

$$\text{magnitude: } \sqrt{\left(\frac{\sqrt{2} - 1}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} = \sqrt{\frac{5}{4} - \frac{\sqrt{2}}{2}}$$

$$\text{phase: } \arctan\left(\frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2} - 1}{2}}\right) = \arctan\left(\frac{\sqrt{2}}{\sqrt{2} - 1}\right)$$

$$\text{gain: } \frac{|N|}{|D|} = \frac{\sqrt{2 + \sqrt{2}}}{\sqrt{\frac{5}{4} - \frac{\sqrt{2}}{2}}}$$

$$\text{phase: } \angle N - \angle D = \arctan\left(\frac{\sqrt{2}}{\sqrt{2} + 2}\right) - \arctan\left(\frac{\sqrt{2}}{\sqrt{2} - 1}\right)$$

$$f = 2 \text{ Hz} \Rightarrow \Omega = 2\pi \frac{2}{8} = \frac{\pi}{2} \Rightarrow z = e^{j\frac{\pi}{2}} = \cos \frac{\pi}{2} + j \sin \frac{\pi}{2} = j$$

$$H(j) = \frac{1 + j}{j - 0.5}$$

$$N = 1 + j \quad |N| = \sqrt{2} \\ \angle N = \arctan(1) = \frac{\pi}{4}$$

$$D = j - 0.5 \quad |D| = \sqrt{\frac{5}{4}} \\ \angle D = \arctan\left(\frac{1}{-0.5}\right) = \arctan(-2)$$

$$|H(j)| = \frac{|N|}{|D|} = \frac{\sqrt{2}}{\sqrt{\frac{5}{4}}} = \sqrt{\frac{8}{5}}$$

$$\angle H(j) = \arctan(1) - \arctan(-2)$$

